Graphs & Transformations

Section 1: Basic Graph Forms & Translations

1 Describe the transformation that maps the curve $y = f(x)$ onto $y = f(x - 3) + 5$.

a. Translation by vector $[3, 5]$

b. Translation by vector $[-3, 5]$

c. Translation by vector $[3, -5]$

d. Translation by vector $[-3, -5]$

2 The curve $y = x^2$ is translated by vector $[-2, -4]$. What is the equation of the resulting curve?

a. $y = (x + 2)^2 - 4$

b. $y = (x - 2)^2 - 4$

c. $y = (x + 2)^2 + 4$

d. $y = x^2 - 2x - 4$

3 Find the coordinates of the turning point of $y = 4 - (x - 1)^2$.

a. $(1, 4)$

b. $(-1, 4)$

c. $(1, -4)$

d. $(4, 1)$

Section 2: Reflections and Stretches

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4 Describe the transformation that maps $y = f(x)$ onto $y = -f(2x)$.

a. Reflection in x-axis followed by horizontal stretch scale factor $1/2$

b. Reflection in y-axis followed by vertical stretch scale factor 2

c. Reflection in x-axis followed by horizontal stretch scale factor 2

d. Reflection in y-axis followed by horizontal stretch scale factor $1/2$

5 The point $(4, 12)$ lies on the curve $y = f(x)$. What are the coordinates of the corresponding point on $y = 3f(x/2)$?

a. $(8, 36)$

b. $(2, 36)$

c. $(8, 4)$

d. $(2, 4)$

6 Which equation represents the reflection of the curve $y = 2^x$ in the y-axis?

a. $y = 2^{-x}$

b. $y = -2^x$

c. $y = -2^{-x}$

d. $y = \log_2(x)$

Section 3: Composite Transformations

7 The graph of $y = x^3$ undergoes a stretch scale factor 2 parallel to the y-axis, followed by a translation by vector $[1, -3]$. Find its equation.

a. $y = 2(x - 1)^3 - 3$

b. $y = (2x - 1)^3 - 3$

c. $y = 2(x + 1)^3 - 3$

d. $y = 2x^3 - 5$

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8 A curve with equation $y = g(x)$ has a local minimum at $(3, -5)$. Where is the corresponding local extremum on the curve $y = -2g(x+1)$?

a. Local maximum at $(2, 10)$

b. Local minimum at $(4, 10)$

c. Local maximum at $(4, -10)$

d. Local minimum at $(2, -10)$

9 Find the asymptotes of the rational function $y = \dfrac{2x+1}{x-3}$.

a. $x = 3$ and $y = 2$

b. $x = -3$ and $y = 2$

c. $x = 3$ and $y = 1$

d. $x = 1/2$ and $y = 3$

Section 4: Curve Sketching & Advanced Analysis

10 Find the coordinates of all intersection points of $y = (x-2)^2(x+1)$ with the coordinate axes.

a. x-intercepts at $(2, 0)$ and $(-1, 0)$; y-intercept at $(0, 4)$

b. x-intercepts at $(-2, 0)$ and $(1, 0)$; y-intercept at $(0, -4)$

c. x-intercept at $(2, 0)$; y-intercept at $(0, 4)$

d. x-intercepts at $(2, 0)$ and $(-1, 0)$; y-intercept at $(0, -2)$

11 The curve $y = \dfrac{a}{x+b}$ passes through $(1, 6)$ and $(4, 2)$. Find the values of a and b.

a. $a = 12, b = 1$

b. $a = 6, b = 0$

c. $a = 18, b = 2$

d. $a = 8, b = 0$

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12 Determine the number of real solutions to the equation $|x^2 - 4| = 3$.

a. 4 solutions

b. 2 solutions

c. 3 solutions

d. 0 solutions

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